

## Ejercicio 1 · Capilaridad — tubo de sección rectangular

Tensión superficial · Contacto adhesión/cohesión · Ascenso capilar

7,5%

✓ Resuelto

a) Obtener la **expresión del ascenso capilar** en un tubo de sección rectangular en que un lado sea el doble del otro (lado corto =  $b$ , lado largo =  $2b$ ).

b) Sabiendo que la tensión superficial del fluido es  $\sigma = 33,8 \text{ dyn/cm}$ , su densidad relativa  $s = 0,95$ , la relación entre fuerzas de adhesión y cohesión es  $7/3$ , y el lado corto de la sección es  $b = 1 \text{ mm}$ , calcular el ascenso capilar.

**Resultados:**  $h = 3\sigma \cos \theta / (\gamma b)$ ;  $h = 9,99 \text{ mm}$

## Datos

| Variable                        | Valor       | SI                            |
|---------------------------------|-------------|-------------------------------|
| Tensión superficial $\sigma$    | 33,8 dyn/cm | 0,0338 N/m                    |
| Densidad relativa $s$           | 0,95        | $\gamma = 9310 \text{ N/m}^3$ |
| $F_{\text{adh}}/F_{\text{coh}}$ | 7/3         | $\tan \theta = 3/7$           |
| Lado corto $b$                  | 1 mm        | 0,001 m                       |

## Fórmulas

## Apartado a) — Expresión del ascenso capilar

Equilibrio de fuerzas en el menisco: la componente vertical de la tensión superficial sobre el perímetro de contacto sostiene el peso de la columna líquida:

$$\sigma \cos \theta \cdot P_m = \gamma \cdot A \cdot h$$

Para sección rectangular  $b \times 2b$ :  $P_m = 2(b + 2b) = 6b$ ,  $A = 2b^2$ :

$$\sigma \cos \theta \cdot 6b = \gamma \cdot 2b^2 \cdot h \quad \Rightarrow \quad h = \frac{3\sigma \cos \theta}{\gamma b}$$

## Ángulo de contacto — modelo físico adhesión/cohesión

En el punto de contacto, la fuerza de adhesión  $A$  actúa perpendicular a la pared (hacia el sólido) y la cohesión  $C$  actúa hacia el interior del fluido formando  $45^\circ$  con la pared. La superficie libre se coloca perpendicular a la resultante de ambas:

$$\tan \theta = \frac{C \sin(45^\circ)}{A - C \cos(45^\circ)}$$

Dividiendo numerador y denominador entre  $C$  y usando  $A/C = 7/3$ :

$$\tan \theta = \frac{\frac{1}{2}}{\frac{A}{C} - \frac{1}{2}} = \frac{0,7071}{\frac{7}{3} - 0,7071} = \frac{0,7071}{2,3333 - 0,7071} = \frac{0,7071}{1,6262} = 0,4348$$

## Teoría e hipótesis

**Capilaridad y tensión superficial:** en la interfaz entre dos fluidos (o fluido y sólido), existe una fuerza por unidad de longitud  $\sigma$  [N/m] denominada tensión superficial. En un tubo capilar, la tensión superficial actúa sobre el perímetro de contacto  $P_m$  con la pared, formando un **ángulo de contacto**  $\theta$  con la vertical.

• Si  $\theta < 90^\circ$ : adhesión > cohesión  $\rightarrow$  el fluido moja la pared  $\rightarrow$  **ascenso capilar** (mercurio:  $\theta > 90^\circ \rightarrow$  descenso capilar).

**Equilibrio del menisco:** la componente vertical de la tensión superficial sostiene el peso de la columna de fluido ascendida:

$$\sigma \cos \theta \cdot P_m = \gamma \cdot A \cdot h \quad \Rightarrow \quad h = \frac{\sigma \cos \theta \cdot P_m}{\gamma \cdot A}$$

Para sección rectangular  $b \times 2b$ :  $P_m = 2(b + 2b) = 6b$ ;  $A = 2b^2 \rightarrow h = \frac{3\sigma \cos \theta}{\gamma b}$ .

**Modelo adhesión/cohesión:** en el punto triple (fluido-sólido-aire), la fuerza de adhesión  $A$  atrae hacia la pared y la cohesión  $C$  actúa a  $45^\circ$  hacia el interior del fluido. La superficie libre se posiciona perpendicular a la resultante, lo que define  $\theta$ :

$$\tan \theta = \frac{C \sin 45^\circ}{A - C \cos 45^\circ}$$

Esta fórmula permite calcular  $\theta$  conocida la relación  $A/C$  de las fuerzas intermoleculares.

## Resolución

### ■ Demostración — Capilaridad en sección rectangular $b \times 2b$

*El profesor pide explícitamente "obtener la expresión": hay que derivarla desde el equilibrio de fuerzas.*

#### ■ Paso 1 — Ángulo de contacto desde adhesión y cohesión

En el punto triple, la fuerza de adhesión  $A$  tira hacia la pared y la cohesión  $C$  a  $45^\circ$  hacia el interior. La superficie libre se posiciona perpendicular a la resultante:

$$\tan \theta = \frac{C \sin 45^\circ}{A - C \cos 45^\circ}$$

Si  $\theta < 90^\circ$  el fluido moja, si  $\theta > 90^\circ$  no moja (mercurio).

#### ■ Paso 2 — Fuerza ascensional en sección rectangular

El menisco en una sección rectangular  $b \times 2b$  tiene perímetro de contacto:

$$P_m = 2(b + 2b) = 6b$$

La componente vertical de la tensión superficial sobre el perímetro es:

$$F_\sigma = \sigma \cos \theta \cdot P_m = 6b \sigma \cos \theta$$

#### ■ Paso 3 — Peso de la columna ascendida

El área de la sección rectangular es  $A = b \cdot 2b = 2b^2$ . El peso de la columna de altura  $h$ :

$$W = \gamma A h = 2\gamma b^2 h$$

#### ■ Paso 4 — Equilibrio estático

El ascenso cesa cuando  $F_\sigma = W$ :

$$6b \sigma \cos \theta = 2\gamma b^2 h \Rightarrow h = \frac{3\sigma \cos \theta}{\gamma b}$$

#### ■ Paso 5 — Comparación con Jurin clásico

Para tubo cilíndrico de diámetro  $D$ :  $h = 4\sigma \cos \theta / (\gamma D)$  (factor 4). Para sección  $b \times 2b$  aparece factor 3 — refleja la mayor relación perímetro/área del rectángulo respecto al círculo de igual diámetro.

*Sin esta demostración el profesor no acepta la fórmula como respuesta válida, según indica el enunciado.*

### 1. ÁNGULO DE CONTACTO $\theta$

**¿Por qué?** El ángulo de contacto  $\theta$  determina si el fluido moja la superficie. Se obtiene de la relación entre adhesión y cohesión. Sin  $\theta$  no se puede aplicar la fórmula de capilaridad correctamente.

$$\tan \theta = 0,4348 \Rightarrow \theta = \arctan(0,4348) = 23,5^\circ \Rightarrow \cos(23,5^\circ) = 0,9171$$

### 2. CONVERSIÓN DE DATOS AL SISTEMA CGS

**¿Por qué?** La fórmula de capilaridad usa  $\sigma$  en dyn/cm,  $\rho$  en g/cm<sup>3</sup> y  $R$  en cm. Convertir todo al mismo sistema antes de operar evita errores de factor 1000.

- Tensión superficial:  $\sigma = 33,8$  dyn/cm
- Lado corto:  $b = 1$  mm = 0,1 cm
- Peso específico:  $\gamma = s \cdot \rho_{\text{agua}} \cdot g = 0,95$  g/cm<sup>3</sup>  $\times$  980 cm/s<sup>2</sup> = 931 dyn/cm<sup>3</sup>

### 3. ASCENSO CAPILAR

**¿Por qué?** El ascenso capilar se obtiene del equilibrio entre la fuerza de tensión superficial ( $2\pi R \sigma \cos \theta$ ) y el peso de la columna de fluido ( $\pi R^2 h \rho g$ ). Despejando  $h$  se obtiene la fórmula de Jurin.

$$h = \frac{3\sigma \cos \theta}{\gamma b} = \frac{3 \times 33,8 \times 0,9171}{931 \times 0,1} = \frac{92,994}{93,1} = 0,9989 \text{ cm}$$

### RESULTADO

$$h \approx \boxed{9,99 \text{ mm}}$$

Fórmula deducida:  $h = 3\sigma \cos \theta / (\gamma b)$ , análoga a la circular ( $h = 4\sigma \cos \theta / (\gamma D)$ ) pero con factor 3 por la geometría rectangular  $b \times 2b$ .

El ángulo de contacto se obtiene del modelo físico vectorial:  $\tan \theta = C \sin 45^\circ / (A - C \cos 45^\circ) = 0,4348 \rightarrow \theta = 23,5^\circ$ . El fluido moja bien la pared, ascenso positivo.

Trabajando en CGS:  $\gamma = 931 \text{ dyn/cm}^3$ ,  $b = 0,1 \text{ cm} \rightarrow h = 0,9989 \text{ cm} \approx 9,99 \text{ mm}$ .

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The study was conducted using a quantitative research design. Data was collected from a sample of 100 participants, and the results were analyzed using statistical software. The findings of the study indicate that there is a significant relationship between the variables being studied.

The results of the study suggest that the research has important implications for the field. Further research is needed to explore the relationship between the variables in more detail.

In conclusion, the study has provided valuable insights into the topic and has contributed to the existing body of knowledge. The findings have important implications for the field and suggest that further research is needed.

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The first part of the paper discusses the importance of the research and the objectives of the study. It then moves on to a literature review, which provides a background on the topic and identifies the gaps in the existing research. The methodology section describes the research design, data collection, and analysis. The results section presents the findings of the study, and the conclusion summarizes the main points and offers suggestions for future research.

The research was conducted in a systematic and rigorous manner, following the principles of good research practice. The data was collected from a representative sample of the population, and the analysis was carried out using appropriate statistical methods. The results of the study are presented in a clear and concise manner, and the conclusions are based on the evidence gathered.

The findings of the study have important implications for the field of research, and they provide valuable insights into the issues being studied. The research also highlights the need for further investigation in this area, and it offers suggestions for how this can be achieved.

In conclusion, the research has shown that there is a need for further investigation in this area, and it has provided valuable insights into the issues being studied. The findings of the study have important implications for the field of research, and they provide valuable insights into the issues being studied.



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The results of the study indicate that there is a significant positive relationship between the variables of interest. This finding is consistent with the previous research in the field. The implications of these findings suggest that the research has practical applications in the field of study.

In conclusion, the study has provided valuable insights into the topic and has contributed to the existing body of knowledge. Further research is needed to explore the topic in more depth and to validate the findings of this study.

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The research was conducted in a systematic and rigorous manner, following the principles of scientific inquiry. The data was collected from a large sample of participants, ensuring the representativeness of the findings. The analysis was performed using advanced statistical techniques, allowing for a detailed examination of the data. The results of the study indicate that there is a significant relationship between the variables under investigation, supporting the hypotheses of the study.

The findings of this study have important implications for the field of research. They provide valuable insights into the underlying mechanisms and processes that govern the phenomenon being studied. These findings can be used to inform the development of new theories and models, as well as to guide the design of future research. The study also highlights the need for further research in this area, as there are still many questions that remain unanswered.

In conclusion, this study has made a significant contribution to the understanding of the subject matter. It has provided a comprehensive overview of the current state of knowledge and identified areas for future research. The findings of the study are robust and reliable, and they have important implications for the field. The study also demonstrates the value of a systematic and rigorous approach to research, and it serves as a model for future studies.

The first part of the paper discusses the importance of understanding the cultural context of the research. It highlights the need for researchers to be sensitive to the values and beliefs of the communities they are studying. This is particularly important in the field of education, where cultural differences can significantly impact learning outcomes.

The second part of the paper focuses on the methodology used in the study. It describes the process of selecting participants, collecting data, and analyzing the results. The authors emphasize the importance of using a mixed-methods approach to capture both quantitative and qualitative data.

The third part of the paper presents the findings of the study. It shows that there are significant differences in learning outcomes between students from different cultural backgrounds. These differences are attributed to a variety of factors, including language barriers, social norms, and access to resources.

The final part of the paper discusses the implications of the findings for education. It suggests that educators should take steps to create a more inclusive learning environment for all students. This can be done by providing additional support for students who are struggling and by incorporating culturally relevant materials into the curriculum.

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In conclusion, the study has shown that the variables are related in a significant way. The findings have important implications for the field, and the study has contributed to the existing knowledge on the topic.

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The implications of the findings suggest that the research has practical applications in the field. Further research is needed to explore the underlying mechanisms and to test the findings in different contexts.

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The findings of the study have several implications for practice and policy. First, the results suggest that the current approach to the issue is not effective. Second, the study highlights the need for further research in this area. Finally, the findings provide valuable insights for the development of new interventions and policies.

In conclusion, the study has provided a comprehensive overview of the research topic. The findings are significant and have important implications for the field. Further research is needed to explore the underlying mechanisms and to develop more effective interventions.

The first of these is the fact that the world is not a uniform whole, but a collection of diverse and often conflicting interests. The second is the fact that the world is not a static entity, but a dynamic one, constantly changing and evolving. The third is the fact that the world is not a simple, linear system, but a complex one, with many interconnected parts and a high degree of uncertainty. The fourth is the fact that the world is not a single, unified entity, but a collection of many different cultures, languages, and peoples. The fifth is the fact that the world is not a single, unified system, but a collection of many different systems, each with its own rules and norms. The sixth is the fact that the world is not a single, unified entity, but a collection of many different entities, each with its own interests and goals. The seventh is the fact that the world is not a single, unified system, but a collection of many different systems, each with its own rules and norms. The eighth is the fact that the world is not a single, unified entity, but a collection of many different entities, each with its own interests and goals. The ninth is the fact that the world is not a single, unified system, but a collection of many different systems, each with its own rules and norms. The tenth is the fact that the world is not a single, unified entity, but a collection of many different entities, each with its own interests and goals.

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The first part of the paper discusses the importance of understanding the cultural context of the research. It highlights the need for researchers to be sensitive to the values and beliefs of the communities they are studying. This is particularly important in the field of education, where cultural differences can significantly impact learning outcomes.

The second part of the paper focuses on the methodology used in the study. It describes the process of selecting participants, collecting data, and analyzing the results. The authors emphasize the importance of using a mixed-methods approach to gain a comprehensive understanding of the research topic.

The third part of the paper presents the findings of the study. It discusses the results of the quantitative data analysis and the insights gained from the qualitative interviews. The authors conclude that there are significant differences in learning outcomes between the two groups, and these differences can be attributed to cultural factors.

The final part of the paper discusses the implications of the findings for future research and practice. It suggests that educators should be aware of the cultural context of their students and tailor their teaching methods accordingly. The authors also recommend further research to explore the underlying reasons for the observed differences.

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The final part of the paper offers recommendations for future research and practice. It suggests that educators should be aware of the cultural context of their students and tailor their teaching methods accordingly. Additionally, it calls for further research to explore the underlying reasons for the observed differences.

